

1. RESEARCH QUESTION

Does stock screening methodology affect the out-of-sample performance of cointegration-based statistical arbitrage?

- Cointegration-based stat-arb requires searching across many candidate asset groups.
- Exhaustive testing is computationally expensive.
- Screening can reduce the search space but may also change the characteristics of the candidates entering the cointegration stage.
- This study compares three screening methodologies (**Full, PCA, and FF3**) and evaluates how the resulting candidate sets affect subsequent portfolio construction and out-of-sample performance.

2. RESEARCH DESIGN

Experimental Framework

- **Universe:** TOPIX 50
- **Validation:** Training: Jan 2022 – Dec 2023 → Test: Jan 2024 – Dec 2025
- **Final OOS:** Training: Jan 2022 – Dec 2025 → Test: Jan 2026 – Aug 2026
- **Candidate groups:** 2–3 assets
- **Selected spreads:** Top 5
- **Common framework:** The same research framework was applied across all screening methods, while the screening methodology and selected trading parameters were varied according to the validation procedure.
- **Parameter selection:** Z-score window and exit threshold were selected separately for each screening methodology based on validation performance and robustness.

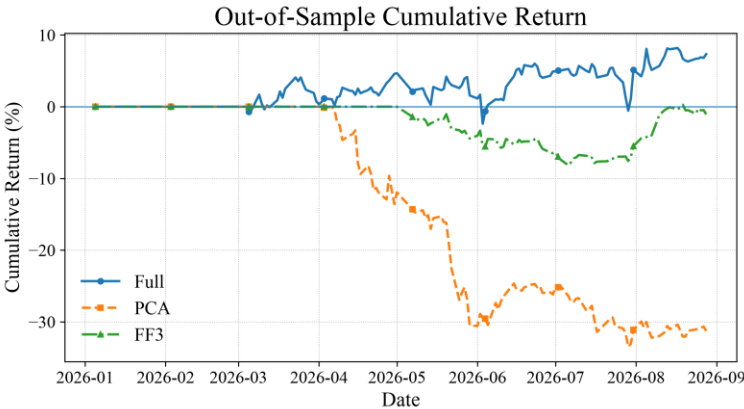
Screening Methods

Method	Representation
Full	Baseline; direct candidate search
PCA	Low-dimensional representation of training-period return patterns
FF3	α + Market + SMB + HML exposures from Fama-French 3-factor regressions

Cointegration & Trading

- **Candidate groups** → cointegration testing → 60-day persistence evaluation → Top 5 selection
- **Spread model:** Kalman filter
- **Entry threshold:** 0.7518, analytically optimized under the Gaussian threshold model
- **Z-score window / Exit threshold:** **Full:** 40 days / 0.25, **PCA:** 60 days / 0.25, **FF3:** 80 days / 0.50

3. EMPIRICAL RESULTS



Selected Cointegration Spreads

Screening Method	Tickers
Full	4063-6146-6301, 5802-6861-7741, 6752-6762-6857, 6857-7741-8411, 6861-9433-9983
PCA	6367-6762-7741, 6501-6503-8591, 6503-7267-8591, 6762-6954-7741, 6762-6981-7741
FF3	2914-4578-9433, 6098-6954-7741, 7267-8053-8058, 7267-8058-8750, 8053-8306-8750

Out-of-sample Metrics

Method	Total Return	Annualized Return	Volatility	Sharpe Ratio	Maximum Drawdown	Win Rate	Number of Trades
Full	0.073	0.118	15.08%	0.770	-6.75%	52.50%	96
PCA	-31.28%	-44.61%	17.37%	-2.84	-33.47%	50.00%	36
FF3	-1.07%	-1.67%	7.46%	-0.23	-8.19%	55.00%	55

4. INTERPRETATION

Key Interpretation

- Screening methodology changes the candidate set and downstream OOS performance.
- Performance differences were driven by the selected spread portfolios.
- The results suggest that candidate generation can materially affect strategy performance.
- No general superiority of any screening method is established.

Key Takeaway

Candidate generation can propagate through cointegration selection and affect trading performance.

5. LIMITATIONS & FUTURE WORK

Limitations

- Single OOS period
- Non-point-in-time universe
- Three screening methods
- Limited parameter sensitivity analysis
- No statistical inference
- Transaction costs not fully incorporated

Future Work

- **Robustness:** Multiple periods and regimes
- **Representation:** Additional screening methods
- **Strategy Design:** Improve candidate ranking
- **Execution:** Incorporate transaction costs
- **Computation:** Scale up cointegration testing
- **Generalization:** Other markets and universes